Reading IMF Programs on Day One

Machine learning on 173 IMF lending programs, 2002 to 2016

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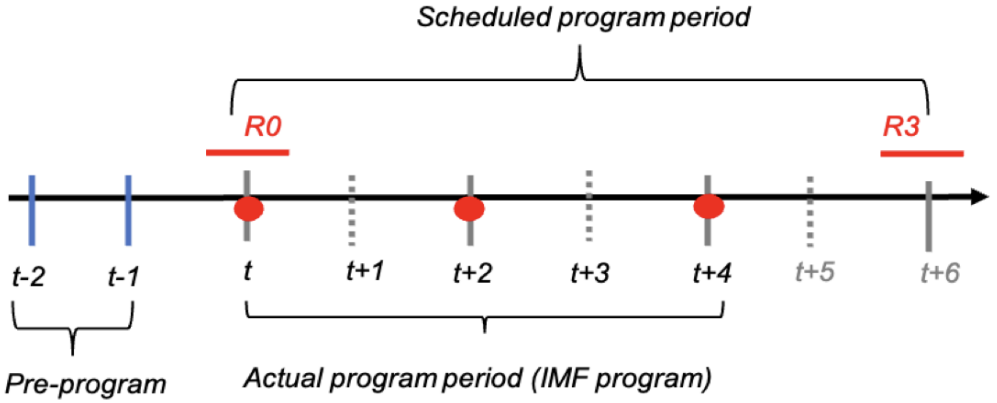
Executive Summary

1. Literature and Research Question
2. Study Goal
3. Data cleaning and feature engineering
4. Models
5. Comparison Across Models
6. Limitations & Conclusion
7. Sources

Literature and research question

- **Financial relevance:** Interruptions can weaken investor confidence and sovereign financing credibility.
- **Political factors:** Political incentives and institutional capacity affect implementation (Ivanova et al., 2003; Joyce, 2006).
- **Conditionality:** Evidence is mixed. The type of structural conditions can matter (Reinsberg et al., 2022).
- **Existing ML research:** Studies predict IMF participation (Agbloyor et al., 2023; Batsuuri et al., 2024).
- **Research gap:** The reviewed studies examine implementation or predict participation. We test whether approval-time information predicts permanent interruption.

Figure Title: Timing of IMF program disbursements



Source: Aiyar, S. and Patnam, M. (2021), *IMF Programs and Financial Flows to Offshore Centers*, IMF Working Paper No. 21/146, Figure 2, p. 8.

- **Can information available at approval predict permanent interruption?**

Study goal: Prediction of IMF programme interruptions at the time of Board approval

Binary classification of programme outcomes using deal design, track record and macroeconomic information available at approval

The objective is predictive accuracy on unseen programmes, not causal explanation of interruptions

- **Y** = 1 if at least one scheduled review was necer completed
  X = 0 otherwise

Y = f(**X**) + ε, where **X** = (X_1, X_2, …, X_p) for p=13 features observed at approval:

  - Deal design
  - Track record
  - Macro conditions

- Purpose: prediction, not inference / the model shall flag the programmes that fail, not explain why. Measured on **test data**, using misclassification error rate.
- Interruption 47%, so a model that always predict "interrupted" already reaches 37% error rate.
- Every interruption weakens market confidence in local policies, so the identification of high risk programmes at approval helps to calibrate conditionality and investors can price sovereign risk more accurately.

Data collection

IMF Financial Data: Commitments	Dates, amounts approved and drawn, access (% of quota), facility type
IMF MONA: Reviews	Review schedule and completion → **Y**
IMF MONA: Conditions, Purchases	Structural conditions, prior actions, front-loading
IMF MONA: Economic indicators	Growth, inflation, current account, as known at approval
IMF WEO (Apr 2026)	Government debt and net lending, % of GDP
World Bank WDI	Fallback for growth and inflation

The sources are linked by country and closest approval date (within ±60 days).

Sample selection criteria

Selection criteria:

1. Facilities with ex-post conditionality and reviews: SBA, EFF, ECF and SCF
2. Approval period 2002-2016: starting when MONA review records **begin**
3. Money at stakte
4. Observed outcome
5. Blends: an EFF and an ECF approved for the same country on the same day count as a single programme approval

Design criteria

Training set: 2002-2013 (n=145)

Test sets: 2014-2016 (n=28)

Tuning: 10-fold CV and LOOCV on the training set.

Missing values: filled with training set medians.

Data from IMF (2026a, 2026b, 2026c) and World Bank (2026); methodology based on Shmueli (2010), He & Garcia (2009) and James et al. (2013).

The data: how we built it and what it looks like

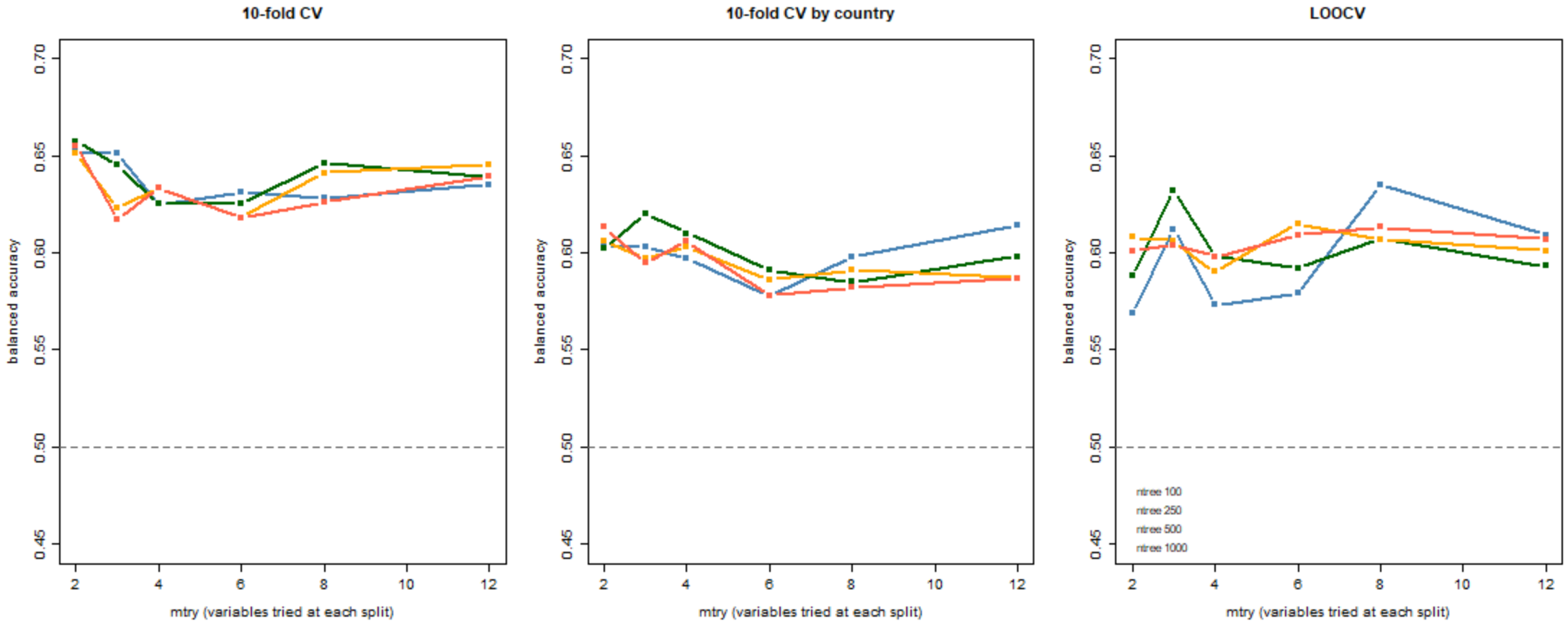
- Five raw files linked without a shared ID: 1,576 arrangements → 173 programmes
- Blends merged into one row
- Outcome built in a loop: reviews scheduled at approval vs. completed
- Only approval-day information: 6 look-ahead variables excluded
- Time split; gaps filled with train medians
- 47% interrupted (82 of 173): train 45%, test 61%
- Interrupted: more scheduled reviews (7.3 vs. 5.9), larger access, fewer concessional and extended ($p \leq 0.004$)
- Conditions, macro, track record: no clear difference
- Test inputs also shifted (more conditions, lower inflation)

The data: Exploration, Inputs, and Output

- SBA 64% and EFF 56% interrupted vs. ECF 38%; total share rising from 38% (2002–07) to 61% (2014–16)
- Weak correlations with output: reviews 0.31, access 0.25, concessional −0.22
- Overlapping inputs (concessional–extended 0.72) and non-linear effects (access: 31 / 52 / 41 / 67% by quartile)
  - Could favour forests and neural nets over a plain linear model
- Output (y-variable): interrupted (0/1)
- 13 inputs: deal terms, conditions, track record, macro
- Rule: known on approval day
- 12 candidate features excluded: 6 look-ahead, 4 duplicates, 1 too many gaps, 1 identifier (country)

Random Forest: tuning

- **Tuning:** 6 mtry × 4 ntree values, scored on balanced accuracy under three schemes.
- **Flat grid:** the lines cross and stay within about 5 points. No pair clearly wins.
- **Choice:** country CV scores lowest because it never sees the test country. Best pair: 250 trees, mtry 3 (BA 0.620).

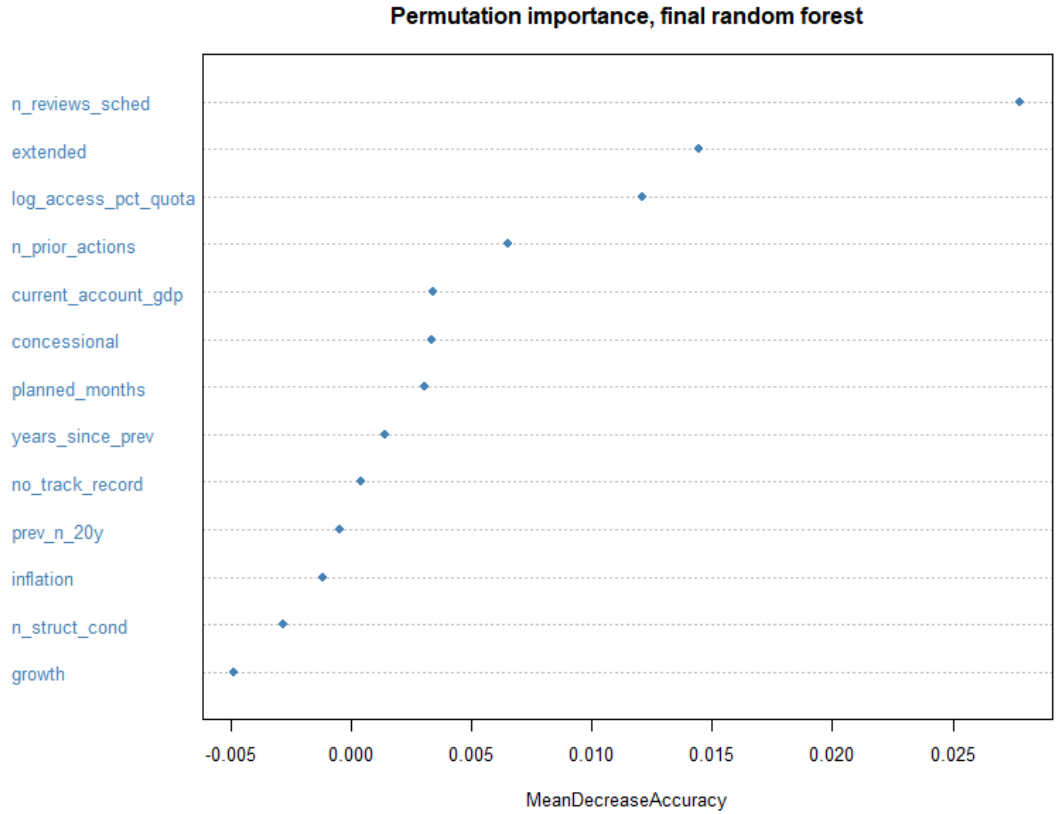


What the forest predicts and with what

	Accuracy	Balanced Accuracy
Dev, out of fold	0,641	0,630
Test, final forest	0,500	0,508
Test, “always interrupted”	0,607	0,500

On the test set, it catches 8/17 interrupted programmes and 6/11 completed ones.

With 28 test programmes the result is mostly noise: over 20 seeds the balanced accuracy ranged from 0.51 to 0.64, and the bootstrap interval is 0.38 to 0.75.

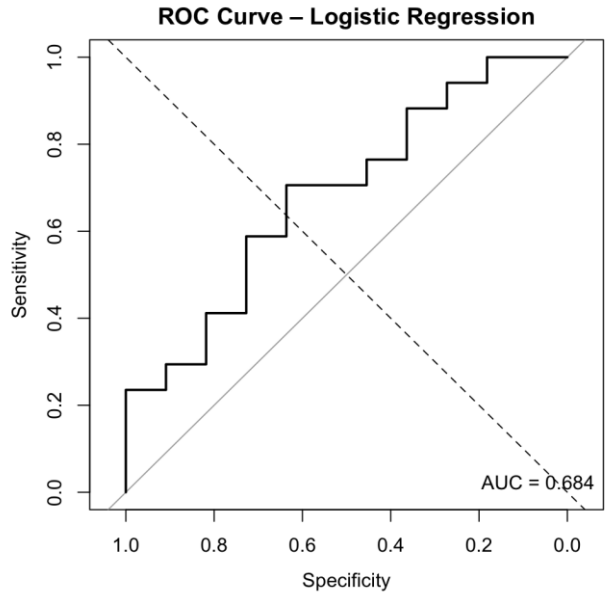


Programme design leads (number of reviews, extended facility, size of the loan). Macro variables add nothing. This is predictive, not causal.

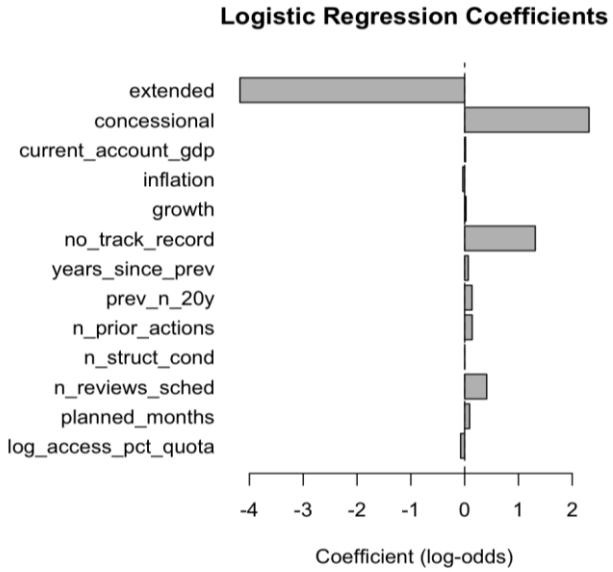
Logistic Regression

Performance

	Log Regression	Random Forest
Accuracy	0.536	0.5
Bal. Accuracy	0.57	0.508
AUC	0.684	0.586



Interpretability



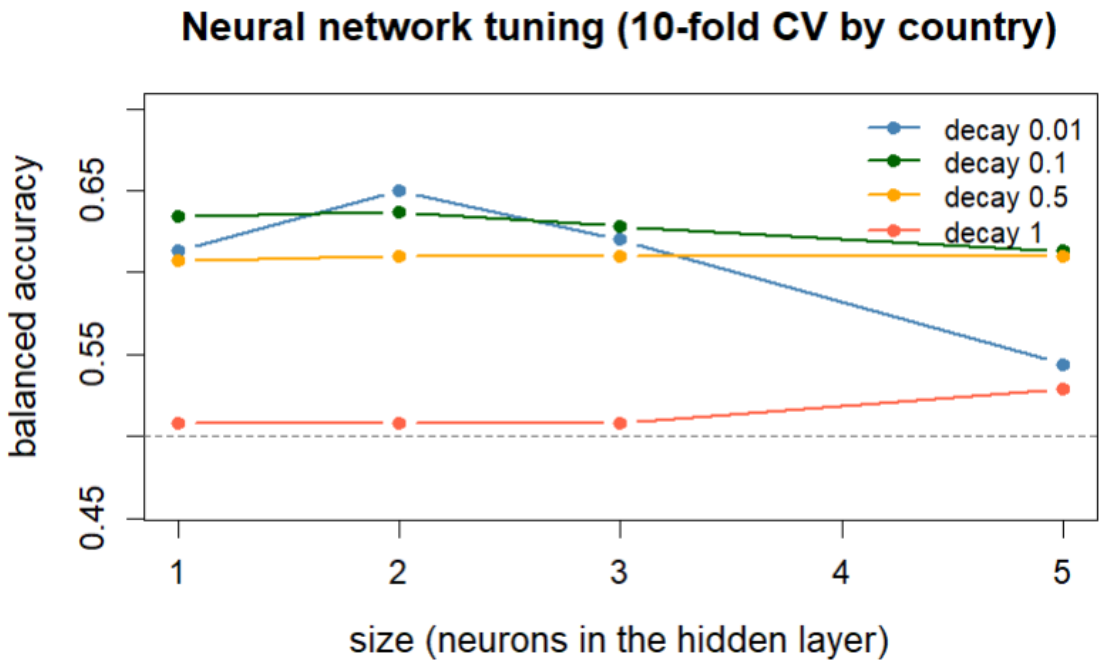
Logistic Regression:

- Clear + / - coefficient's effect
- Easier to interpret and explain
- Simpler model Structure
- Less flexible

Random Forest:

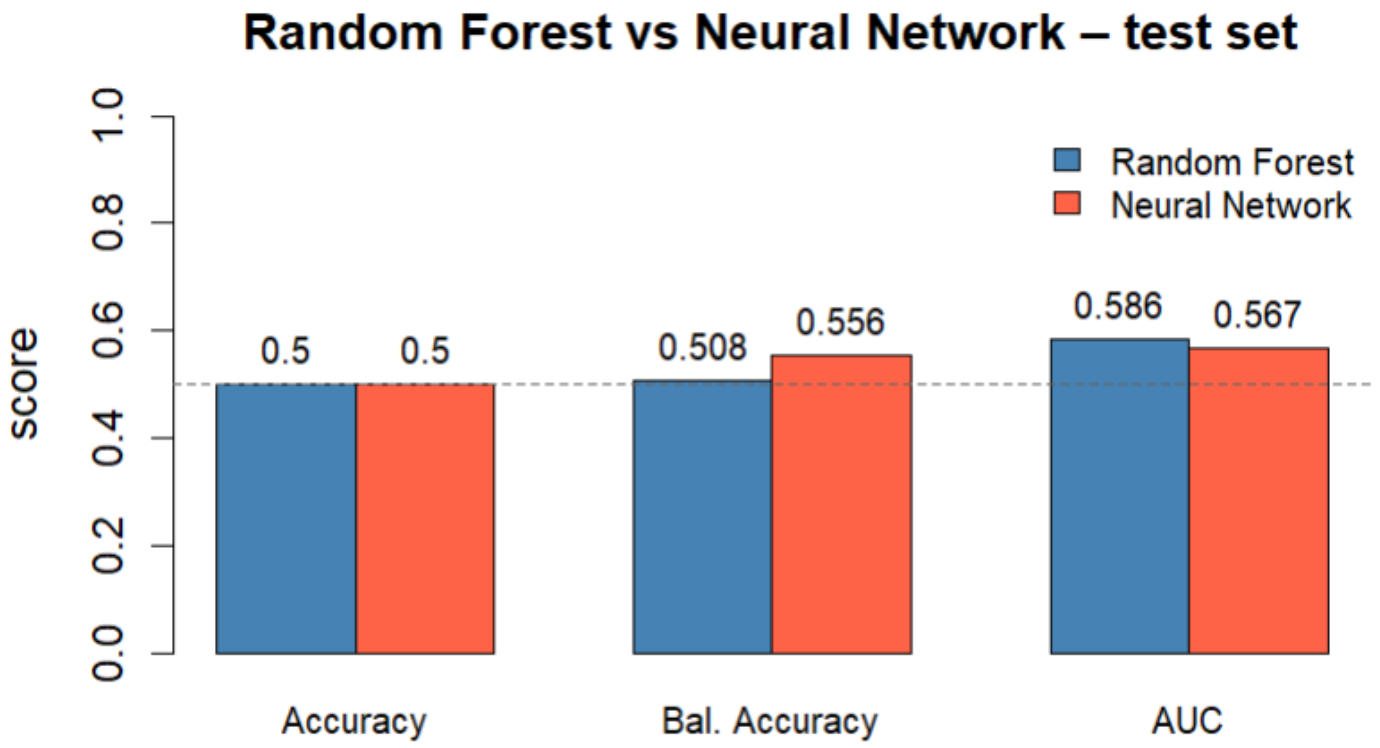
- Captures non-linear relationships
- More flexible
- Harder to Interpret

Neural Network



- **Small is enough:** the best score comes from 2 neurons with a light penalty
- **Overfitting shows up:** with a light penalty, going from 2 to 5 neurons drops the score to about 0.54.
- **Too much penalty kills it:** with decay 1, the score sits at 0.50 for every size.

Model Comparison



Neither RF nor NN is reliably better than the other

With only 145 training programmes, a neural network can't show the advantage it's built for

Limitations and Conclusion

Limitations:

- **Small sample:** 145 programmes to train on and 28 to test, of which only 11 completed. One extra correct call moves balanced accuracy by about 0.05.
- **Single time split:** the share of interrupted programmes jumps from 45% in dev to 61% in test, so the test period isn't like the training period.
- **Coarse inputs:** only approval-day data. Politics, shocks after approval and the government's commitment aren't in the model

Conclusion:

- **Weak signal:** about 0.63 balanced accuracy out of fold, but no reliable edge over chance on the test set.
- **Deal design predicts best:** the number of reviews, the facility type and the loan size matter most. Macro inputs add nothing.
- **Metric matters:** with unequal classes, plain accuracy rewards a blind "always interrupted" rule. Balanced accuracy shows the real skill.

Sources

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